Practice Test 08 — Answer Key

Module 12: Muscular System

Part A: Multiple Choice

1. **B** — Skeletal muscles are attached to bones and produce voluntary movement; contraction also generates heat.
2. **B** — When muscles contract, ATP is used and heat is released as a byproduct, helping maintain body temperature.
3. **B** — Whole muscle → fascicle (bundle of fibers) → muscle fiber (single cell) → myofibril (contractile strand within the fiber).
4. **C** — The sarcomere is the repeating unit between two Z-lines; it is the basic contractile unit of striated muscle.
5. **B** — Thick filaments are composed of the motor protein myosin.
6. **C** — Thin filaments are primarily actin, with regulatory proteins troponin and tropomyosin.
7. **C** — Endomysium wraps each individual muscle fiber; perimysium wraps fascicles; epimysium wraps the whole muscle.
8. **B** — The thin (actin) filaments slide inward along the thick (myosin) filaments, pulling the Z-lines closer together and shortening the sarcomere.
9. **C** — A new ATP molecule must bind to the myosin head to release it from actin, allowing the cycle to continue or the muscle to relax.
10. **B** — Calcium binds to troponin, causing tropomyosin to shift and expose the myosin-binding sites on actin.
11. **C** — Acetylcholine (ACh) is the neurotransmitter released by motor neurons at the neuromuscular junction.
12. **B** — ACh binding opens ion channels, depolarizing the muscle membrane and triggering an action potential that leads to contraction.
13. **B** — A motor unit = one motor neuron + all the muscle fibers it innervates. Recruiting more motor units increases force.

14. **C** — Tetanus is a smooth, sustained contraction from rapid, repeated stimulation where individual twitches fuse.
15. **B** — In isotonic contractions the muscle shortens (or lengthens under load) and moves something. In isometric, the muscle generates force but doesn't change length.
16. **B** — Fatigue results from ATP depletion, accumulation of lactic acid and other metabolic waste, and impaired calcium release.
17. **B** — Slow-twitch fibers are fatigue-resistant, rich in mitochondria and myoglobin, ideal for endurance activities and postural muscles.
18. **C** — Fast-twitch fibers generate more force and contract faster, but rely on anaerobic metabolism and fatigue more quickly.

Part B: Fill in the Blank

1. **Myosin** (thick); **Actin** (thin)
2. **Acetylcholine** (ACh)
3. **Calcium** (Ca^{2+}) ions; **Troponin**
4. **Isometric**; **Isotonic**
5. **Epimysium** (outermost), **Perimysium** (middle), **Endomysium** (innermost)

Part C: Short Answer

1. **Cross-Bridge Cycle (4 steps):**
    - **Step 1 — Myosin binds actin:** The energized (cocked) myosin head attaches to an exposed binding site on actin, forming a cross-bridge.
    - **Step 2 — Power stroke:** The myosin head pivots, pulling the thin filament toward the center of the sarcomere. ADP and phosphate are released.
    - **Step 3 — ATP binds:** A new ATP molecule binds to the myosin head, causing it to detach from actin.
    - **Step 4 — Cocking:** ATP is hydrolyzed (→ ADP + P), and the energy re-cocks the myosin head into the high-energy position, ready to bind again.

2. **Why Both ATP and Calcium Are Needed:**

- **Calcium** is the "on switch": it removes the tropomyosin block from actin so myosin can bind. Without calcium, no cross-bridges form.
- **ATP** is the energy source: it powers the cocking of the myosin head and causes the myosin head to release from actin. Without ATP, myosin remains locked onto actin (rigor).

3. **Slow-Twitch vs. Fast-Twitch:**

- **Slow-twitch (Type I):** Slower contraction speed, high endurance, many mitochondria, high myoglobin (red). Best for: marathon running, posture, walking.
- **Fast-twitch (Type II):** Rapid contraction, powerful, fewer mitochondria, fatigue quickly. Best for: sprinting, jumping, heavy lifting.
- Everyone has a mix of both; training can shift some fiber characteristics.

4. **Isometric Contraction:**

- This is **isometric**: the bicep is generating force to hold the book steady, but the muscle is not changing length (no movement at the joint). In isotonic contractions, the muscle would shorten or lengthen while moving the load.